

Chapter 47: Application of Machine Learning in Improved Diagnosis, Treatment and Disease Monitoring of People Living with HIV (PLWH)

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Abstract

Human immuno-deficiency virus (HIV) causing acquired immuno-deficiency syndrome (AIDS) is still a threatening disease in human beings. Therefore, various strategies of artificial intelligence and machine learning prediction studies on HIV testing, CD4 counts, viral load, antiretroviral therapy (ART) follow up, risk factors, drug resistance mutations, treatment outcomes could be helpful in better diagnosis of people living with HIV (PLWH). The monitoring of disease progression and treatment success in HIV study using artificial intelligence and machine learning based prediction is very essential. To focus on scaling up HIV treatment activities on antiretroviral therapy (ART), the assessment of viral load and CD4 counts using machine learning algorithms predicts better treatment outcomes for PLWH. Currently, manual-based HIV testing index creating many challenges, including time consuming, errors on report preparation and ART follow up, expensive to operate in a daily routine test. The use of the artificial intelligence in machine learning could help to predict and visualize the diagnosis index of the PLWH. The implication of standard machine learning algorithms such as K-nearest Neighbor (KNN), Decision Trees (DT), Gaussian Naive Bayes (GNB), Logistic Regression (LR), Support Vector Machine (SVM), eXtreme Gradient Boosting (XGB), Random Forest (RF) could predict the CD4 and viral load counts, ART status for diagnosis of people living with HIV. The use of artificial intelligence-based machine learning algorithm models can assist the decision makers for a better diagnosis of people living with HIV.

Keywords: HIV, AIDS, AI, ML, ART, PLWH

1. Introduction

Acquired immuno-deficiency syndrome (AIDS) of humans is caused by human immuno-deficiency virus-1 (HIV-1) and human immuno-deficiency virus-2 (HIV-2). The origin and evolution of these HIV were observed through genetic changes that occurred as simian immuno-deficiency viruses (SIVs) crossed from monkeys to apes and then transfer its genetic changes from apes to humans provides a new

insight to examine a future zoonotic risk. Host-specific restriction factors played important role in transmission of HIV in individuals of human beings. Human to human passage of various HIV strains would be expected to increase the chance of adaptations in human host [1]. AIDS is a human viral infectious disease caused by positive sense single stranded (ss) RNA circulated in HIV-1 and HIV-2. The genome size of HIV is around 9.1 kb encoding 15 distinct proteins. The control of HIV infection in people living with HIV could possible through multiple antiretroviral therapy belongs to different classes of drugs such as nucleoside reverse transcriptase inhibitors (NRTIs), non-nucleoside reverse transcriptase inhibitors (NNRTIs), Protease inhibitors (PIs), Integrase inhibitors, entry inhibitors. The appearance of several resistant strains towards drugs in people living with HIV impacts the immune system [2-3]. HIV test results through based on antigen and antibody interactions and detections in several generations of diagnostic assay was performed [4].

HIV self-testing (HIVST) is addressed by world health organization (WHO) for its sensitivity and specificity for a testing option for the delivery of pre-exposure prophylaxis (PrEP). HIVST remains a better diagnostic tool in community-based programme, HIV service delivery programme, initiation of antiretroviral therapy. HIVST is fruitful for a correct clinical decision for the health care providers. The use of AI vision computer technology shown a promise tool for providing the additional quality assurance of HIV testing excluding false negative test interpretations committed by human end users [5]. The application of artificial intelligence (AI) including machine learning (ML) is a useful tool in the field of HIV prevention. In these countries like U.S, Denmark, Eastern Africa, ML approaches are being utilized in improving the delivery of preexposure prophylaxis (PrEP) in health care settings. In the area of real time HIV reduction study, the other applications of ML have been implemented with smart phone collected and social media data, HIV Sero disclosure data and chatbots for HIV education study data. The in-lodge of AI and ML strategies for HIV prevention is a mode of implementation science approach for qualitative assessments of end users [6]. The monitoring of disease progression and treatment success of the people living with HIV adults' patients enrolled for antiretroviral therapy (ART) care could be assessed for viral load and CD4 counts using machine learning algorithms. Xtreme gradient boosting (XGB), Random Forest (RF) models had the highest accuracy on HIV-1 viral load testing. Gradient boosting (GB) model had the highest accuracy on prediction of CD4 counts of HIV-1 patients [7]. Deep learning models was utilized using Ethiopian demography and health survey (EDHS) data sets for prediction of HIV test results. The use of the machine learning model predicts accuracy of the individuals HIV test status [8]. The identification of HIV predictors and predicting person at high risk of infection is very important. The concern of HIV status through the targeted screening of risk predictors could improve the health care. A potential XGBoost algorithm of machine learning techniques was identified as a potential model for HIV predictors screening in sub-Saharan Africa [9]. Antiretroviral therapy is a better option transformed HIV patients from a fatal illness to a chronic disease. Drug adherence of HIV individuals is very important for successful management of HIV treatment. Interruption in treatment (IIT) on ART could impact the life style of the people living with HIV. The use of high performing machine learning models could improve the routinely collected service delivery data at a risk of interruption in treatment. The use of ML models in interruption in treatment improve the routine management health system through increased cost-effective ness and improved patient outcomes [10].

Therefore, we generalized our review through collection of study of various articles on the use of artificial intelligence and machine learning for a better diagnosis, treatment and monitoring of people living with HIV.

2. Utilization of AI based ML study in HIV testing of PLWH

HIV/AIDS is a major cause of death and could be preventable by making strategies for proper diagnosis and treatment in early prediction. So, there is need for a major tool that can help to stop the early prediction of the spreading the disease. Infectious diseases are caused by several infectious agents such as Fungai, bacteria, viruses etc. and its early detection, prevention and management can be possible by machine learning (ML) methods [11]. The reduction of the HIV infection and incident related mortality in both high and low resource settings is very essential. The most important challenges of the study include the use of effective tools for the collection of behavioural and sociodemographic data of routine HIV testing programme is a successful challenge in HIV prevention and control in a low resource setting. The integration of machine learning and artificial intelligence algorithms capture the behavioural and sociodemographic data of HIV testing program and has proven the risk of HIV infection [12]. Counselling is an important structure in HIV testing and evaluate the scenario of social attributes and risk factors for HIV transmission among different population level. Evaluation of innovative HIV testing services with web based and the use of artificial intelligence chatbot increased the real time counselling among the men who have sex with men [13]. HIV is still a threatening disease in the society. The manual-based HIV index testing create many challenges such as time consuming, expensive and errors. Machine learning models such as Random Forest (RF), XG Boost, and Artificial neural networks (ANN) algorithms in HIV index testing predict and visualize the best results. ML models assist the decision makers to adopt a suitable intervention strategy towards ending HIV/AIDS [14]. Several machine learning models and algorithms such as Random Forest, X-treme gradient booster, Support vector machine, logistic regression, deep learning, decision tree exhibiting higher accuracy in predicting HIV testing compared to traditional tests. The integration of various ML models improves the efficiency of HIV testing services and optimize resource allocation in a resource setting programme [15]. ML methods utilized the textual data from electronic health records (EHRs) of HIV /AIDS patients and predicts the better opportunity of early diagnosis of HIV patients [16]. Regular and timely testing for sexually transmitted infections is very important for the HIV and sexually transmitted infected (STI) people. Several ML models such as naïve bayes, gradient boosting machine, vector machine, K-nearest neighbour, random forest, XG Boost, multilayer perceptron were established and useful to predict timely clinic attendance for HIV/AIDS patients. Generally, the XG boost model performed the best predicting model for timely clinically attendee and the elastic net regression model performed the best predicting model for HIV/STI testing. These two models are useful in healthcare and follow up services [17].

3. Utilization of AI based ML study in Risk prediction of PLWH

The age of an individual has the highest impact and more aged people in both male and female sexes have a higher probability of infection. People with low education have a greater risk of HIV infection in both sexes. People with low CD4 count have a more opportunistic infections with Tuberculosis. Literacy and Urbanity as a strong predictor of HIV acquisition and disease spreading in both sexes. Identifying HIV predictors and predicting persons at high risk of infections through XGBoost algorithms along with SHAP plots of socio-behavioural driven data of male and female population [9]. The most important challenges in the HIV risk prediction study uses various tools for the collection of behavioural, sociodemographic and clinical profile data, the appropriate contextualization and local validation of

models. The successful integration of ML systems in PrEP and routine HIV testing services in Low Middle Income (LMIC) Settings. ML applied to HIV risk prediction made a potential significant contribution towards ending the HIV epidemic. When there is a persistently high HIV incident rates, accurate and granular risk prediction of HIV incident is critical and is a threatening on ongoing transmission to determine on recent progress on HIV burden. ML and AI algorithms captures the various complex interactions data has proven to be effective in predicting the risk of HIV infection in both high and low resources settings. The evaluation of various machine learning algorithms with the help of emerging tools such as shapely additive explanations must help to decompose and understand of various ML models in the context of HIV [12]. Data mining and machine learning is an important study in HIV research. Through machine learning, statistically risk analysis of HIV infection, using various models for HIV infection risk prediction, preventive measures study based on HIV risk of infection could be possible, various diseases of AIDS patients and various machine learning methods are used to predict the risk of HIV/AIDS in a population [18]. Socio-behavioural data is an important predictive data for individuals at high risk of infections in a targeted screening. Identification of HIV predictors improve the health care of high-risk HIV infected population. The potential use of XGBoost algorithm with socio-behavioural driven data substantially identified the HIV predictors and predict the high-risk HIV infection in a targeted screening population [9]. The machine learning algorithms are commonly used in clinical data to predict the risk of acquiring HIV, gonorrhoea, chlamydia, syphilis etc. The HIV/STI risk prediction tools offer useful method for sexual behavioural interventions study to reduce the future HIV/STI risk. EHR based machine learning could predict risk of HIV/STI over the next 12 months. The incorporation of ML model-based risk assessment tools in clinics helps to promote future HIV/STI testing or identification of HIV PrEP individuals or reduction in HIV/STI HIV risk in early interventions [19].

4. ML based study in treatment of antiretroviral therapy for PLWH

The artificial intelligence is applied in the HIV treatment regimens to revolutionized the optimization of therapy. AI also transforming the treatment strategy on focusing the individual therapy on tailor made regimens on patients need and also enhancing the treatment outcomes. AI could optimize the HIV treatment through utilization of computational techniques and advanced algorithms in a patient data including viral load, CD4 counts, and treatment history of patients [20]. High performing ML models are used to identify the HIV risk in the routine testing data. ML could improve the targeted interventions through various models of care in treatment and treatment interruption and predict the cost-effective ness improved patient's outcome. ML models such as boosting tree and extreme gradient boosting could predict the future of interruption in treatment [10]. ML methods were applied to the electronic medical records data of loss to follow up adult HIV positive individuals those who were newly enrolled for antiretroviral therapy. The collected data were utilized through six supervised ML classifiers- J48 decision tree, K- nearest neighbours, logistic regression, naïve bayes, random forest etc. ML model predicts the assessment of future risk of LFTU in HIV care in a low resource setting [21]. In Yunnan province, China, the investigation of clinical monitoring indices of HIV/AIDS patients receiving antiretroviral therapy (ART) could be a best possible through various ML models. Three machine learning models such as support vector machine (SVM), Random Forest (RF) and Multilayer perceptron (MLP) were constructed for the clinical indicators such as base line and follow up results of routine blood testing data of HIV/AIDS patients under ART. The use of three ML Models such as MLP, RF and SVM with clinical indicators better predicted the immune function and recuperation of HIV/AIDS patients and

could be helpful in better guiding in clinical treatment [22]. HIV attacks body's immune system permanently. ART helps to reduce the spread of virus and increase the life expectancy of the people infected in HIV. The ML model Random Forest with an accuracy of 90% predict the effect of ART on people living with HIV [23]. ML models could help the HIV programmes implemented in targeted intervention study to improve the quality of care, to find out the effective early intervention, insight into optimal resource allocation and program running and sustainability. The AdaBoost model measure and predicts the risk factors, prior behaviour, visit history and geographic factors and demographic features of the PLWH on ART [24]. AI with personalized medicine managed the HIV transmission by enhancing diagnosis, disease monitoring and treatment optimization. The use of ML models, deep neural networks, multi omics data analysis during prognostication, continuation of ART, and early detection of drug resistance mutations is the major implementation part during the management of PLHIV. AI driven various models could analyse vast genomic, proteomic and clinical data sets of treatment strategy, prediction of disease progression and therapy failure. AI powered diagnostic tools such as deep learning imaging, natural language processing and improved screening accuracy could predict the management in care of PLWH. AI driven approaches hold a promise for effective and efficient in specialized medicine for HIV treatment and care [25]. AI and ML models could be utilized for prediction of CD4 and viral load status of people living with HIV on ART. The XGB and RF models had the highest accuracy on viral load and the GB model had the highest accuracy on prediction of CD4 counts [7]. ML models such as neural network, logistic regression tree-based models predict the risk of patients in loss to follow up of ART [26]. HIV virologic failure during treatment is a key health global challenge. ML model nanogram based scoring system predict the incidence and determinants of virologic failure in ART among people living with HIV [27].

5. ML based study in HIV prevention of PLWH

The implementation of AI/ML could be an important solution for improving the effectiveness of HIV/AIDS prevention program. An HIV prevention chatbot could aid encourage self-testing, personalized treatment strategies and prevention messaging. AI/ML applications implementation could create a great challenge in HIV prevention [28]. The integration of AI and ML in HIV research including diagnostic innovations, therapeutic strategies, viral mechanisms and prevention mechanisms is a great challenge in future HIV management. The integration of AI/ML model in various health applications in HIV/AIDS work like an advance study and gives the future direction in management of HIV/AIDS [29]. The development, discovery and integration of AI/ML could have a potential impact on HIV cure. Various ML models are implemented for analysis of the age, gender, race, concurrent diagnosis, percentage linked to care within three months, prevalence of PLHIV/AIDS, percentage of viral suppression, CD4 counts, Drug adherence, Treatment options, death rate etc. The integration of AI/ML enhances the individual treatment plans for HIV cure and management [30]. AI/ML models have a strong potential for PrEP delivery, integration of ML in smartphones with collected and social media data promote real time HIV reduction. The use of AI /ML approach in HIV/AIDS research could help ending the epidemic [6]. AI and ML could enhance HIV awareness, rapid treatment, improving early HIV infection detection strategy for prevention. AI played a pivotal role in revolutionizing health care and has a great potential in HIV prevention intervention strategies. AI and ML technologies create a vision for HIV elimination in management of health care [31].

6. Conclusion

The integration of AI/ML studies in HIV/AIDS predicting drug resistance from genomic data, optimizing antiretroviral therapy and CD4 counts, viral load etc. It also identifies the risk populations using digital data and social media through text analysis and image platform. The integration of AI/ML identify the potential candidates for pre-exposure prophylaxis (PrEP), and also powers chatbots for HIV education, promoting self-testing and play important role in prevention to establish a health care setting.

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